Foundational Research & Advances in GHZ State (Greenberger-Horne-Zeilinger)

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for GHZ State (Greenberger-Horne-Zeilinger) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Three entangled particles that are so tightly linked that a single measurement on each particle disproves Einstein's local realism with 100% certainty in ONE shot."

3. Mathematical Formulation & Unitary Rule

```
|\text{GHZ}_n\rangle = \frac{|0\rangle^{\otimes n} + |1\rangle^{\otimes n}}{\sqrt{2}}, \quad (
```

Maximally entangled state violating local realism deterministically.

4. Step-by-Step Circuit Sequence

Hadamard gate on qubit 0 followed by a cascade of CNOT gates: CX(0,1), CX(1,2)...

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(4)
qc.h(0)
qc.cx(0, 1)
qc.cx(1, 2)
qc.cx(2, 3)
# State is (|0000> + |1111>)/sqrt(2)
print(qc.draw(output='text'))
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
Requires 2^(n-1) classical correla[illegible]	Prepared in O(n) CNOT depth (or O(log n) with tree topology)	Foundational

7. Key Applications & Future Research Directions

- Quantum metrology achieving the Heisenberg limit (Delta theta ~ 1/N)
- Quantum Byzantine agreement and multi-party secret sharing
- Benchmarking coherence across physical quantum processors

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Fundamentals pipelines.